

Use Case: Pay Order (VietQR)

Use case code: UC003

Brief Description: This use case involves the integration of payment using the VietQR method, allowing customers to pay for physical media products (books, newspapers, CDs, DVDs) via a QR code.

Actors

- Customer: The user purchasing physical media.
- VietQR Service: The external system providing the API for QR generation and transaction sync.

Preconditions

- The system has calculated the current price of products.
- The system has created connection to VietQR.
- The customer has placed an order for the products in their cart with delivery information.

Basic Flow of Events

1. The customer selects QR code as the payment method for their order.
2. The system calls the VietQR Payment API and send information(See Table A).
3. VietQR Payment generate a dynamic QR code based on the order details and send to the system(See Table B).
4. The system displays the generated QR code to the customer.
5. The customer completes the payment using the QR code.
6. VietQR system handle the transaction and send transaction information to system.
7. The system returns the payment transaction result and information to the calling use case ("Place Order").

Alternative flows

No	Location	Condition	Action	Resume location
1.	At Step 2	If the system cannot connect to the VietQR system	▪ The system displays error message.	Use case ends
2.	At Step 4	If QR code is invalid or expired	▪ The system displays the message "Payment failed or overdue". ▪ The system takes the customer back to the order page.	Use case ends
3.	At Any Step	If the customer wants to cancel the payment transaction(e.g. customer wants to change payment method)	▪ The system displays the message "Cancelled Transaction". ▪ The system takes the customer back to the order page	Use case ends

Input data

Table B

No	Data fields	Description	Mandatory	Valid condition	Example
1.	QR Link	The URL contains a QR code image automatically generated by the VietQR system.	Yes	Must be a valid HTTPS URL	https://img.vietqr.io/image/mb-1234567890-compact.png?amount=500000&addInfo=PAY98765

Output data

Table A

No	Data fields	Description	Display format	Example
1.	BankCode	Recipient's bank	Required, String	MB
2.	BankAccount	Recipient's bank account	Required, String (Numeric characters only)	99991112345
3.	UserBankName	Recipient name	Required, String	NGO DUC HUY
4.	TotalAmount	The actual amount of the order	Required, Integer, greater than 0	150.000
5.	Content	The transfer details will be visible to the customer.	Maximum 19 characters, written continuously without spaces or accents, and no special characters.	ORD12345
6.	OrderId	Transaction code for system to manage.	Required, String, Unique	ORD987654321

Postconditions

- The payment transaction is authorized and confirmed by the VietQR system. Transaction details (including Transaction ID, Transaction Datetime, and success status) are returned to the Use Case (Place Order) to continue the order recording flow.
- The transaction was cancelled, the QR code expired, or the payment failed. A failure/cancellation status is returned to the Use Case Place Order for the system to process the next step.